AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I128*

Team Name: *Logic Loom*

Domain: *Food security,Agriculture and Rural transformation*

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Presentation Outline

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Problem Statement

Core problem:
Food wastage and loss occurs in different forms from the farmer's surplus agricultural produce to the excess food preparation in the hotels and canteens. Farmers may struggle to find the buyers for surplus goods to be sold while Hotels may expect higher sales which in turn rather leads to excess food as wastage.

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Poor demand forecasting, fragmented coordination, transport difficulties and delayed surplus detection contiribute to food wastage.

Problem Understanding

- This problem is experienced by:
  Farmers: Face difficulty in matching the surplus agricultural produce with the suitable buyer at the right time.
  Hotels and canteens: uncertain meal demand can lead to overproduction or shortages of meal.
  Food recovery Organizations : Face difficulty in finding and collecting suitable surplus within a certain time.

- Major causes:
  1. Unpredictable supply and demand.
  2. Poor coordination between suppliers and buyers.
  3. Delayed identification of surplus.
  4. Lack of systematic waste tracking.

- Consequences:
  Financial loss, insufficient resources, avoidable food waste, missed opportunities to redistribute suitable surplus

Existing Gaps & Challenges

- Existing gap 1:
  Lack of prediction and prevention: Existing solutions often focus on redirecting the excess food rather than focusing on the actual cause that is preventing the surplus by predicting the amount of food to be produced. So there is a gap in connecting demand prediction and waste prevention.

- Existing gap 2:
  Delayed surplus identification and coordination: Farmers may struggle to find suitable buyers for their surplus agricultural produce in the right time. Poor coordination, transportation constrains and delays can cause food to remain unsold or becoming waste.

- Main challenges:
  To predict potential surplus early and coordinate time reactions to prevent avoidable food loss and waste.

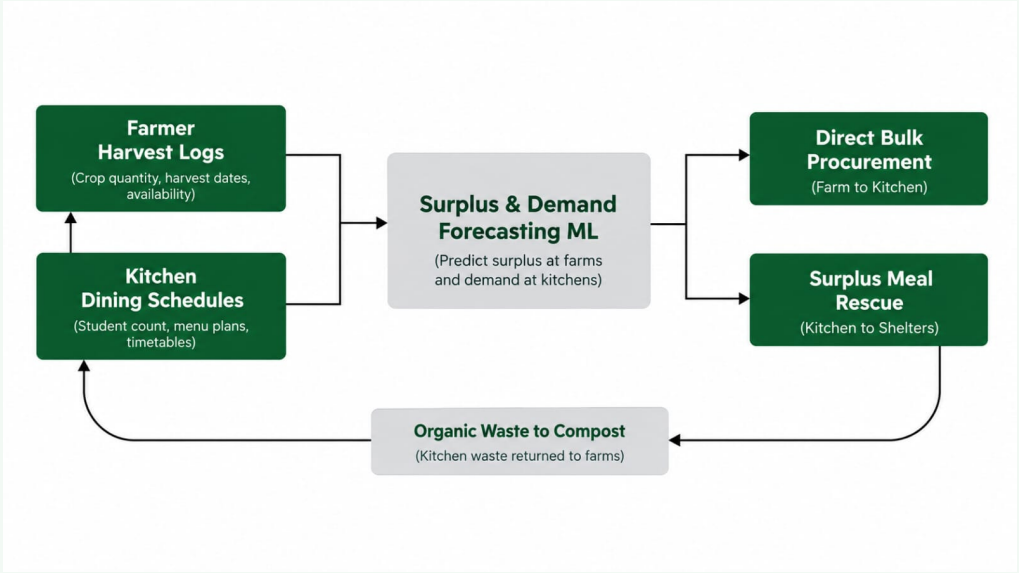
Our Proposed Solution

- Core idea :
  an AI powered platform that helps food loss and waste by connecting agricultural suppliers , food businesses, buyers and food recover organisations.

- How the solution works :
  1. Predict : estimate food demand and identify potential surplus using available data.
  2. prevent:recommend suitable purchasing quantities and meal preparation.
  3. connect: connects farmers with suitable buyers based on the quantity price and the location.
  4. Recover: connect edible surplus with suitable food recovery organizations for timely collections.

- Main values delivered:
  1. Helps farmers find buyer for surplus produce.
  2. Helps hotels and canteens reduce over purchasing and excess food preparation.
  3. improves coordination between food suppliers buyers and recovery organisations.

Technical Approach



Tech Stack

Frontend: [HTML,CSS,JavaScript]

Backend: [Python with Fast API]

Database: [PostgreSQL]

AI/ML, APIs, or Hardware: [Python,Pandas,Scikit-learn]

Innovation & Uniqueness

- Our Software combines demand prediction, food waste prevention, agricultural buyer mapping and surplus recovery in one integrated platform.
- It addresses food waste at both the agricultural supply stage and food preparation stage where early planning can help avoid unnecessary surplus of food.
- Our platform links early surplus identification with buyer matching and recovery coordination rather than treating these activities as completely separate processes.

Feasibility & Scalability

- **Feasibility:**

  1. Resources: web application, database, sample food demand and supply data and abasic AI and ML tools.
  2. development: build user dash boards, demand predictions, supplier buyer matching and surplus tracking.
  3. Implementation: start with small pilate involving hostel or canteen and few local suppliers.

- **Scalability:**

  1. begin with one campus or local area
  2. expand to more farmers ,FPOs, hotels and food recovery organisations.
  3. extend platform to multiple locations by adding users suppliers and regional data

- **Risks:**

  1. Insuffiecient data : Start with sample data and improve data as real data becomes available.
  2. unexpected demand changes: Allow staffs to update demand and available stock in real time.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Hostel demand forcasting:
  1. A Hostel kitchen estimates the number of meals needed.
  2. Our platform analyses past meal counts and no demand patterns.
  3. The system recommends a preparation quantity.
  4. actual meals served and left overs are recorded

- Use case 2: Agricultural surplus matching:
  1. An FPO lists available vegetables with quantity price and location.
  2. Our platform identifies potential buyers.
  3. Buyers confirm quantities and prices.
  4. delivery and order status are tracked.

  Note: Additional slides may be included only for this section.

Use Cases (compulsorily 3 use cases must be there)

- Use case 3: Unexpected demand and surplus recovery
  1.unexpected meeting creates a sudden increase in meal demand
  2.stop update the additional head count
  3.our platform checks available ingredients,preparation time and kitchen capacity
  4.it recommends feasible actions or alerts staff if a shortage is likely
  5.if suitable surplus remains,the platform helps timely collection by an appropriate recipient

Impacts and Benefits

- Benefits:
  1. Farmers: Better buyer opportunities.
  2. Food businesses: reduce excess purchase and preparation.
  3. Food recovery organisations: Easier surplus food collection.

- Overall Impact:
  1. Social: Supports food redistribution
  2. Economic: helps reduce food related losses.
  3. Environmental: Reduces avoidable food wastages.

- Measurable outcome :
  1. Less food discarded
  2. Imporved demand prediction accuracy
  3. More surplus food sold or recovered.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 12	Target 12.3	responsible consumption and production	Reduce food waste and food loses across supply chain
SDG 2	Target 2.1	Zero hunger	Support excess to safe and nutricious food through suitable surplus recovery.

Individual Member Contribution

Member	Contribution or responsibility
Badhma sree vignesh S	Coordinated the team, identified the problem statement, defined the solution concept and organised the presentation.
Darshan B	Proposed AI-based demand forecasting and surplus prediction, defined matching logic and identified use cases and expected impact.
Prithivram B	Developed the proposed application architecture, selected suitable technologies and planned the database and user interface.
Mahathi S	Researched food loss and waste challenges, identified user needs and designed the SmartFood application workflow.

Yukti ID

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